



BY AMETECs

MR-01

SmartEPT — Consolidated Engineering Report, Five Sessions

Everything built, fixed and decided · 11–12 August 2026 · Product + Central + Live

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1. How To Read This Report

This document consolidates **five working sessions of 11–12 August 2026** into one record for Ejaz and the Ametecs team. For every piece of work it answers four questions: **What** was done, **Why** it was needed (the trigger — a live error, a report from Ejaz, or a design gap), **How** it works now, and **the Reason** behind the chosen approach. Its companion, **MR-02**, is the consolidated test-case workbook the team executes to verify all of it.

One reassurance up front, because the last session looked dramatic: **no approved feature or price was changed at any point**. The final session repaired the *testing machinery* (broken silently since 19-July) and fixed two small genuine bugs the revived tests caught. Every behaviour your clients see is either exactly as it was, or exactly as decided with Ejaz in these sessions.

State at the time of writing: **all work is committed locally on Ejaz's PC. Nothing new has been pushed or deployed to live yet**. Section 10 is the single deploy runbook.

2. The Five Sessions At A Glance

#	Session	Focus	Key commits
S1	11-Aug (day)	Product login root-cause + client-wise SMTP + forgot-password · Live VPS health triage · admin delete/cleanup · progressive lifetime pricing	Product <code>22ed12e</code> · Central <code>e6f5839</code> <code>d8279a5</code> <code>73888fc</code> <code>72c0059</code> <code>2d9d8a6</code> <code>385836a</code> <code>a03f228</code>
S2	12-Aug (day)	Technical SEO series + GA4 root cause · 3 live product bugs · documentation pack	Central <code>3ce729f</code> <code>706b642</code> <code>44f71d2</code> <code>2fa3fab</code> · Product <code>1f6be35</code>
S3	12-Aug (evening)	OTP resend cooldown · licence-URL auto-config · branded-URL API passthrough · agent-500 root fix · payment history	Product <code>389b8a0</code> <code>979ccba</code> <code>1c12fb0</code> <code>1c4cdcb</code> · Central <code>a055ed3</code> <code>558417c</code> <code>b6be8c8</code>
S4	12-Aug (night)	Per-tenant licensing + Tenants (HOST) screen	Product <code>1fcb7f7</code> · Central <code>76ddf09</code>
S5	12-Aug (late night)	Licence change flows (upgrades/downgrades/.lic) · test-system recovery	Central <code>face8e7</code> · Product <code>01174cc</code> <code>aef81df</code> <code>408785c</code> <code>2ca7dec</code> <code>da11247</code> · Central <code>75edea3</code> <code>bd7df9f</code> <code>d28b56c</code>

3. Session 1 — 11 August: Product Access, Email & Money Foundations

3.1 The login root cause (Product 22ed12e). *What:* every seeded login (super@smartept.io and all demo users) was fixed so `password` actually works. *Why:* no seeded account had EVER been able to sign in — local or live. *How/Reason:* the `User` model declares `password => 'hashed'` (Laravel hashes on assignment) AND the seeder also called `Hash::make()` — so passwords were hashed **twice** and could never match. The seeder now passes the plain value once. Standing lesson recorded for all projects: with a `hashed` cast, never pre-hash; and `where()->update()` bypasses casts — use `forceFill()->save()` for passwords. A tinker one-liner was given to unlock the live super account after deploy.

3.2 Client-wise SMTP (same commit). *What:* every company can carry its own SMTP (host/port/user/encrypted password/from), edited on the Ops → Email/SMTP card; a **global relay** (super-only) sits behind it. *Why:* clients want alerts/OTPs sent from their own mail identity; and the install needs one reliable fallback. *How:* `MailService::resolveMailer(companyId)` resolves **company SMTP** → **global settings SMTP** → `.env` — every mail in the product (alerts, credentials, OTPs) routes through it; test buttons verify a config before saving. *Reason for the 3-level chain:* a client's broken SMTP must never silence system mail — fallback is automatic.

3.3 Forgot-password with email OTP (same commit). *What:* "Forgot password?" on the console login card, all roles including employees. *Why:* password resets previously required an admin. *How:* request-OTP endpoint (throttled 5/min, neutral response so emails can't be probed, tenant-slug locked) → 6-digit hashed OTP, 10-minute expiry, 5 attempts → reset with a new min-8 password. Delivery uses the company-SMTP resolution above. Fully audit-logged.

3.4 Live VPS health triage (with Ejaz, one issue at a time). Four findings on the live Central server: the **licence signing key** was missing (existed only locally, gitignored by design) — Ejaz overrode: keys committed `2d9d8a6`, so the repo must stay **private forever**; the **hosted-console link** was unconfigured — provisioning secrets generated and set in Central Settings by Ejaz; **Interakt WhatsApp** key ready to configure; and the **hourly dunning job was crashing** — `tenants.status` was an ENUM without `'purged'`, so the purge step died with error 1265 every hour (**ENUM strike three**). Fixed by converting the column to VARCHAR (`385836a`) — per the standing Ametecs rule born here: *the application layer owns value sets; columns are VARCHAR, never ENUM.*

3.5 Admin delete / garbage cleanup (Central 73888fc , 72c0059). *What:* super-only Delete for quotes/unpaid orders (only when nothing was ever received and no licence hangs off them) and invoices (with an FY-gap warning, pre-launch only); money-bearing orders are **never** deletable; no Edit anywhere. The /buy coupon box also hides unless a live coupon exists. *Why:* test garbage from the pre-launch period cluttered real books. *Reason for the guards:* the ledger's integrity outranks tidiness — anything that ever touched money is permanent.

3.6 Progressive lifetime pricing (Central `e6f5839` + `d8279a5`). *What:* On-Premise (perpetual) price bands became **milestone prices with straight-line interpolation** between them — one engine (`PricingService::calculateLifetimeLicencePrice`) used identically by `/buy`, the portal calculator, auth pages and the landing widget; counts above the last milestone return **Custom Quote** (price NULL — ₹0 is impossible by construction); AMC prices at 18% of the interpolated figure. *Why:* fixed band jumps made 26 users cost the same as 40. *Reason:* clients pay proportionally; Ametecs edits only the milestones. A 64-case acceptance matrix (`PerpetualProgressivePricingTest`) locks the arithmetic.

3.7 Also that day: the TM/® sweep (`a03f228` — house rule: © only, nothing is trademarked yet) and **Razorpay verified smartept.com** — live keys active.

4. Session 2 — 12 August (day): SEO, Live Bugs, Documentation

4.1 Technical SEO series (Central `3ce729f` `706b642` `44f71d2` `2fa3fab`). Five strictly-scoped tasks for Google Search Console: **(T1)** `sitemap.xml` became an explicit HTTPS allow-list of the 7 public pages — `canonicalBase()` force-upgrades `http` → `https` so a wrong live `APP_URL` can never leak `http` URLs again; **(T2)** self-referencing canonicals on all 7 public pages; **(T3)** workflow pages (`client login/signup/portal`, `/buy`, `/pay`, `/cms-preview`) send `X-Robots-Tag: noindex, follow` — deliberately **without** `robots.txt Disallow`, because Google must crawl a page to see its `noindex`; **(T4)** manual Nginx steps (server-only, still unverified) to `301 www` → `non-www` in a single hop; **(T5)** the GA4 mystery root-caused: the Landing Editor's publish pipeline rendered from an **empty** `landing_sections` table (import never run on live), so publish silently never rewrote `landing.html` — publish now **self-heals** by auto-importing, and `trackingTags()` injects GA4/GTM/Pixel/Ads on the legal pages and `/buy` too. Editor semantics fixed in words: **Save stores, Publish applies**. Ejaz confirmed the live landing + editor working.

4.2 Three live product bugs from Ejaz's log paste (Product `1f6be35`). **(a)** Attendance derivation crashed daily with "lessThan() on string": three timestamp columns added on 23-Jul were never added to the model's casts — fixed; lesson: *new timestamp column ⇒ cast in the same commit*. Derivation self-heals on its next run — no data was corrupted, writes had simply failed. **(b) ENUM strike four:** the shared Rules screen sends one `action_on_blocked` value to both policy tables whose ENUMs disagreed (one allowed CLOSE, the other BLOCK) — both columns → `VARCHAR(20)` (`2026_08_12_000200`). **(c)** `restoreArchive` undefined-key — already fixed locally, just undeployed.

4.3 Documentation pack: SR-01 session report, SR-02 77-case suite, SR-03 bug-fix report, BF-01 business-flow document (9 flowcharts), BF-02 testing guide (14 journeys / 68 steps + Excel tracker) — all in `docs\SmartEPT-Session-2026-08-12\` .

5. Session 3 — 12 August (evening): Three Live Reports Run Down

5.1 "OTP not received" → configuration, plus a real improvement (Product 389b8a0). The live server's global SMTP was unset and `.env` mail was `log` — OTPs were "sent" into a logfile. Config action for Ejaz (set global SMTP as super). Code improvement shipped alongside: a **60-second resend cooldown** with a counting-down "Resend code" button, matching a new server-side per-account throttle.

5.2 "Licence UNCONFIGURED" → self-configuring Central URL (Product 1c12fb0 , Central b6be8c8). The product needed `SMARTEPT_LICENSE_URL` in `.env` — missing on live (and not even in `.env.example`). Now `LicenseClient::baseUrl()` resolves: `.env` override → the `license_central_url` Setting **pushed by Central at provisioning** → the default `https://smartept.com` . Reason: a client server must never require a `.env` edit to phone home.

5.3 Branded-URL agents (Product 979ccba). Cloud clients get ONE address — `admin.smartept.com/<slug>` — and agents configured with it now work out of the box: anything under `/<slug>/api/...` 308-redirects (method + body preserved) to the real `/api/...` . Verified live.

5.4 The agent HTTP-500 root cause (Product 1c4cdcb) — an important lesson. A client PC failed sign-in with 500. The log showed a duplicate-key on `employee_devices.device_uuid` : the PC had registered earlier under a **different company** (Ejaz's own tests), and the `BelongsToCompany` global scope made that row *invisible* to the lookup — so the code inserted a duplicate instead of updating. Fix: lookups on globally-unique columns use `withoutGlobalScopes()` ; the device row re-points to the new company with a `DEVICE_COMPANY_REASSIGNED` audit entry, and the old owner's agent token dies. **Lesson (all projects): any lookup on a globally-unique column must bypass tenant scopes, or cross-tenant rows become invisible and re-insert.**

5.5 Payment History (Central a055ed3) + coupon Edit (558417c). The client modal gained a full **Payment History ledger** — every `order_payments` row with method/reference and a "Total received" line — so partial/NEFT collections are auditable at a glance. Coupons became editable in place.

6. Session 4 — 12 August (night): Per-Tenant Licensing & The Tenants Screen

6.1 Why. Ejaz, signed in as the product's Super Admin, saw **Khan Incorporation's licence** presented as the server's own. Root cause: `installation_licenses` was a single row per install — correct for client-hosted servers, wrong for the shared cloud install hosting many

companies. Every tenant ran on Khan's 25 seats, whole-box storage was reported against Khan, and one tenant's expiry would have stopped everyone.

6.2 The concept, locked with Ejaz. Central = **business super admin** (orders, billing, licences — the sole commercial truth). Product Super Admin = **host/operator** (cross-tenant operations, owns no licence). Each cloud tenant (`deployment_model = AMETECs_SAAS`) carries its **own** licence row. Formal decisions: shared install is the permanent cloud model; build licensing + Tenants screen together; auto-move Khan's key in the migration; Tenants menu at the top (HOST group); non-cloud companies stay on the install-level slot.

6.3 What was built (Product `1fcb7f7` , 14 files). `installation_licenses.company_id` (nullable, unique — NULL = install row, so client-hosted behaviour is byte-identical); a single decision point `InstallationLicense::governing(company)` used by the agent gate, seat activate/release, the daily validator and storage reporting — every tenant phones home with its own key and reports **only its own** storage; `/api/provision` accepts `licence_key` so **Central delivers the key at provisioning** (Central `76ddf09`) — a cloud client never sees a licence screen; the Licence page is scope-aware per role; and the new **HOST → Tenants** screen (Super Admin only) shows per tenant: licence health, devices vs seats, users, employees, **storage used vs quota with a % bar** (10-minute cache), last agent activity, branded-console links and a per-tenant Validate button. Commercial data deliberately stays in Central — one source of truth.

6.4 The one live side-effect to manage: after migration the install slot is empty (Khan's key moves out) — non-cloud companies on the shared install (including Ametecs' own /admin company) start a fresh 7-day evaluation. Ejaz must either issue an internal licence for the install slot or ask for the operator company to be exempted — within 7 days of the live migration.

7. Session 5 — 12 August (late night): Licence Change Flows + The Test-System Recovery

7.1 Licence change flows (Central `face8e7`). Ejaz's report: "upgrade / downgrade / seats — how, and how to bill it — not found anywhere; and how to re-issue an updated .lic." The survey found one hidden flow (the portal's cloud pro-rata card, 6-Aug) and three genuine gaps. Policies locked: perpetual = **upgrades only, never reduced, never refunded**; perpetual upgrade price = **difference of interpolated lifetime prices** (worked example on Ejaz's bands: 15 → 30 users = ₹10,000 + ₹1,800 GST), AMC re-bases automatically; cloud reductions happen **only at renewal**, schedulable by client or admin, never below devices in use. Built in one commit: the admin **"Upgrade..."** button (billed order + GST invoice + pay link — replaces silent limit edits), `licences.renewal_device_limit` with the full schedule → bill-reduced → apply-and-clear cycle, the portal's perpetual-upgrade and reduce-at-renewal cards with a why-not note, and the portal **".lic download"** button that re-signs **only with the stored fingerprint** — clients can refresh their file after an upgrade but can never move machines without the admin's investigated Shift-machine action.

7.2 The test-system recovery — what actually happened, in plain words. When Ejaz first ran the new tests, *everything* failed with zero assertions. The investigation peeled three layers, none of them touching product behaviour: **(1)** `config/database.php` had hardcoded `'default' => 'mysql'` since 19-July (itself a fix against a rogue env var) — which also silently defeated the test system's SQLite, so **no product test had truly run since 19-July**; it now yields to SQLite only under `APP_ENV=testing` (`408785c`). **(2)** `phpunit's` env entries don't override machine-level variables — a stray `DB_CONNECTION` on the PC hijacked test runs; now forced (`aef81df` , Central `75edea3`). **(3)** Four old migrations used raw MySQL `ALTER TABLE ... MODIFY` with no driver guard and crashed SQLite — guarded (`01174cc` , Central `bd7df9f`); guards are no-ops on live MySQL where those migrations already ran.

7.3 What the revived tests immediately caught — proof the safety net matters. Two genuine bugs: the provisioning endpoint 500'd when the optional `timezone` / `admin_name` fields were omitted (`2ca7dec`), and the meeting-reminder query used MySQL-only `DATE_SUB` syntax (`da11247` — works identically on MySQL, now portable). And 21 **stale test expectations** — all encoding July's prices and July's behaviour, superseded by decisions Ejaz approved since (setup fee covers 30 devices; Pricing v2 dropped the $\times 1.5$ cloud multiplier; the `smartept` plan exists and trials provision on it; the single-session guard; employee delete = archive with a 200 response; offline-first licence validation; required meeting modes). The tests were updated to the approved current behaviour (`d28b56c` , `da11247`) — **the engine was right; the July tests were out of date.**

7.4 Final scores: Central 56 / 56 passed (342 assertions) — the first fully green Central run, covering smoke, portal, financial flows, GST, licence change flows and progressive pricing. Product at last run: 130+ passing with the final fixes committed — the team's first MR-02 execution includes re-running the full suites to confirm the end state.

8. Consolidated Commit Register

Product (smartept), oldest first: `22ed12e` login fix + SMTP + forgot-password · `1f6be35` 3 live bugs · `389b8a0` OTP resend cooldown · `979ccba` slug API passthrough · `1c12fb0` licence URL auto-config · `1c4cdcb` agent-500 unscoped device lookup · `1fcb7f7` per-tenant licensing + Tenants screen · `01174cc` migration sqlite guard · `aef81df` phpunit force env · `408785c` test-DB root fix · `2ca7dec` provision optional-fields fix · `da11247` reminder query driver-aware + suite reconciliation.

Central (smartept-central), oldest first: `2d9d8a6` signing keys (repo private!) · `385836a` tenants.status VARCHAR · `73888fc` admin delete · `72c0059` coupon box · `a03f228` TM sweep · `e6f5839` + `d8279a5` progressive lifetime pricing · `3ce729f` `706b642` `44f71d2` `2fa3fab` SEO series · `a055ed3` payment history · `558417c` coupon edit · `b6be8c8` provisioner announces central_url · `76ddf09` provisioner sends licence key · `face8e7` licence change flows · `75edea3` phpunit force env · `bd7df9f` migration sqlite guards · `d28b56c` suite reconciliation.

9. Decisions Log (all five sessions)

Decision	Ruling
Signing keys in repo	Ejaz override: committed — repo stays PRIVATE forever
Columns with fixed value-sets	VARCHAR always, never ENUM (four strikes)
Trademark symbols	© only; no ™/® anywhere until registered
Perpetual pricing	Milestone bands + interpolation; above bands = Custom Quote; ₹0 impossible
Cloud hosting model	ONE shared install (admin.smartept.com) for all cloud tenants — permanent
Licensing on shared install	Per-tenant licence rows; Central = business super admin; product super = host/operator
Khan's key	Auto-moved by migration to Khan's own tenant row
Non-cloud companies on shared install	Keep the install-level licence slot
Perpetual changes after purchase	Upgrades only — never reduced, never refunded; price = interpolated difference; AMC re-bases
Cloud seat reductions	At renewal only; client or admin schedules; never below devices in use
.lic self-service	Stored fingerprint only; machine moves stay admin-controlled
Deferred	Storage top-up self-service (next) · pause-subscription (needs policy) · plan switching (needs a 2nd plan) · AMC billed flow

10. The Deploy Runbook (nothing is live yet — do in this order)

Step 1 — local final check (both repos): product `php artisan test` and central `php artisan test` — expect green (Central 56/56 confirmed; product re-run after the final commits). Laragon Stop All → Start All.

Step 2 — push both repos.

Step 3 — live product (admin.smartept.com): `git pull && php artisan migrate --force && php artisan config:clear && php artisan route:clear && php artisan view:clear`, reload php8.3-fpm. **Step 4 — live Central** (smartept.com): `git pull && php artisan migrate --force && php artisan view:clear && php artisan config:clear && php artisan route:clear`.

Step 5 — live configuration & checks, in order: (1) Tenants screen: Khan ACTIVE on scope "own" (else press Validate; else re-Save the tenant in Central — provisioning now pushes the key). (2) **Decision within 7 days:** licence the now-empty install slot or exempt the operator company. (3) Set **global SMTP** (Ops → Email/SMTP as super) then test forgot-password end-to-end. (4) Tinker-unlock super@smartept.io and **change every seeded password**. (5) Apply the **Nginx www → non-www 301** and submit sitemap.xml in GSC; confirm live `.env` `APP_URL=https://smartept.com`. (6) One Khan agent signs in and syncs.

Step 6 — hand MR-02 to the team. Sign-off rule: every P1 passes ⇒ the whole 11–12 Aug body of work is approved on live; any P1 failure stops and comes to Ejaz immediately.

11. Risk Register & Open Items

1. **Install-slot 7-day evaluation** after the Khan key move — needs the Step-5 decision (highest-attention item).
2. **Nginx www 301 + GSC submission** — server-side, still unverified.
3. Repo privacy is now a hard requirement (signing keys committed).
4. Ejaz's PC carries a stray machine-level `DB_CONNECTION` variable — harmless now (tests force their own), optional cleanup.
5. Central `_to_delete/tmp_obj_*` files bloating git history (Ejaz-side commit `374c0d6`) — suggest `.gitignore` + `git rm -r --cached` at leisure.
6. Deferred features (Section 9) are agreed, not scheduled.
7. Interakt WhatsApp key was pasted in chat on 11-Aug — regenerate if in doubt.

12. About MR-02 — The Team's Test Suite

MR-02 (Excel workbook + PDF, delivered with this report) consolidates **all testable work of the five sessions** into ~130 cases across three parts: **Part 1** — Sessions 1–3 features (login/SMTP/forgot-password, live-bug verifications, OTP cooldown, branded URLs, licence auto-config, payment history, SEO spot-checks); **Part 2** — per-tenant licensing & the Tenants screen (58 cases); **Part 3** — licence change flows (39 cases). Each case has preconditions, steps, expected result, a priority (P1 = go-live gating) and result-tracking columns. The Summary sheet self-computes progress. Testers work through Part 1 on live after deploy, Parts 2–3 first on local Laragon and then the P1s again on live.

Prepared by Claude (Cowork) for Ametecs Pvt Ltd — internal engineering record, sessions of 11–12 August 2026. SmartEPT Central remains the single commercial source of truth.